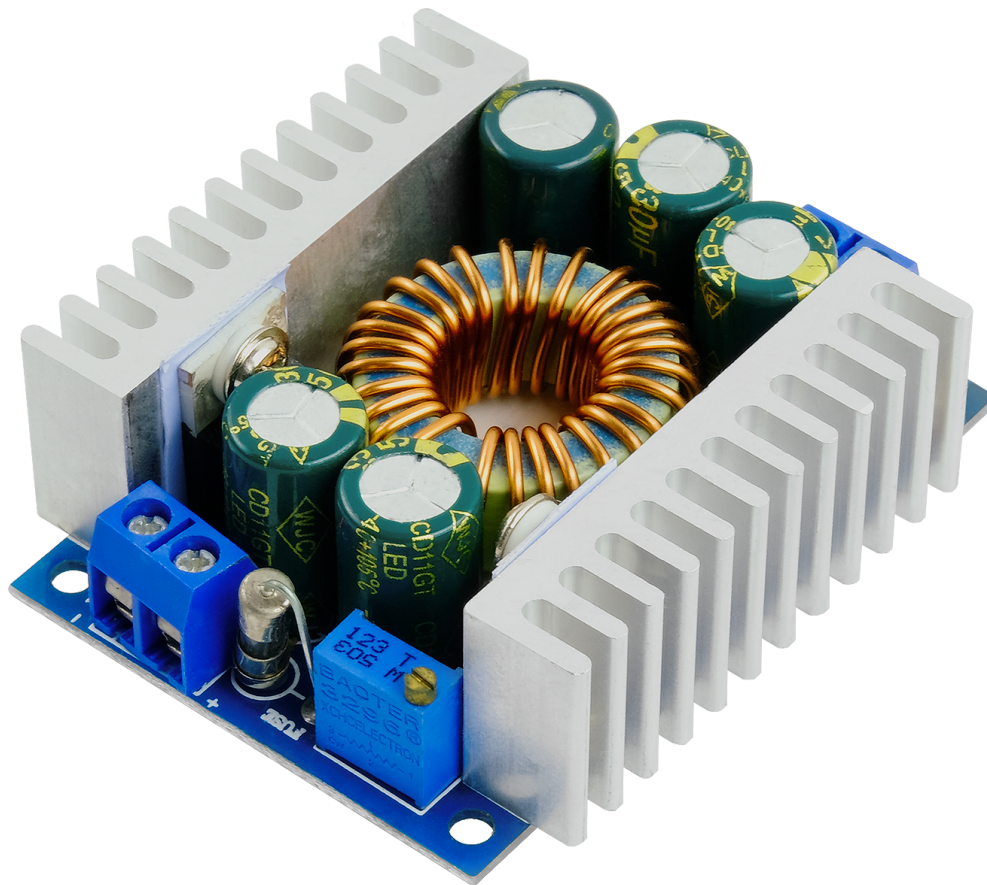


XL-4016E1

eBook for

XL-4016E1 Step-down Converter



Areas of application

The power supply is intended for use in electronic circuits, laboratories, educational institutions and development environments. It is used to safely provide direct current for a variety of low-voltage applications.

Required knowledge and skills

Using the power supply requires basic knowledge of electrical engineering. Users should be able to safely assemble and operate electrical circuits and take the necessary safety precautions.

Operating conditions

The DC power supply should only be operated in dry, well-ventilated indoor areas. The ambient temperature should be between 0°C and 40°C and the relative humidity should not exceed 80%. For safe operation, the power supply must be installed in a suitable housing

Environmental conditions

The power supply must be used in an environment that is free from moisture and direct sunlight. It should be operated away from flammable materials and liquids to ensure safety.

Intended Use

The product is designed for use in educational, research and development environments. It is used to develop, program and prototype electronic projects and applications. The product is not intended as a finished consumer product, but rather as a tool for technically savvy users, including engineers, developers, researchers and students.

Improper foreseeable use

The product is not suitable for industrial use or safety-relevant applications. Use of the product in medical devices or for aviation and space travel purposes is not permitted

disposal

Do not discard with household waste! Your product is according to the European one Directive on waste electrical and electronic equipment to be disposed of in an environmentally friendly manner. The valuable raw materials contained therein can be recycled become. The application of this directive contributes to environmental and health protection. Use the collection point set up by your municipality to return and Recycling of old electrical and electronic devices. WEEE Reg. No.: DE 62624346

electrostatic discharge

Attention: Electrostatic discharges can damage the product. Note: Ground yourself before touching the product, such as by wearing an anti-static wrist strap or touching a grounded metal surface.

safety instructions

Attention: Improper disposal of electronic components can endanger the environment and health. Note: Dispose of electronic components in accordance with local regulations and use appropriate recycling options. Attention: If damaged, electronic components can release harmful substances. Note: Avoid contact with damaged components and dispose of them properly. Attention: Improper disposal of electronic components can endanger the environment and health. Note: Dispose of electronic components in accordance with local regulations and use appropriate recycling options. Attention: If damaged, electronic components can release harmful substances. Note: Avoid contact with damaged components and dispose of them properly. Attention: Mechanical shocks can cause damage and malfunction. Note: Avoid mechanical shock and stress on the power supply. Caution: Inadequate fastening can lead to accidents. Note: Make sure the power adapter is securely and firmly mounted to prevent accidental movement or falls. Attention: Damaged components can affect the functionality of the entire system. Note: Check the power supply regularly for visible damage and replace defective parts immediately. Caution: Short circuits can result in electric shock and fire. Note: Avoid short circuits by using insulated tools and appropriate protective covers for electrical connections. Caution: Overvoltage can damage the power supply and connected devices. Note: Strictly adhere to the power supply's specified voltage and current limits. Caution: Direct contact with live parts can result in injury. Note: Make sure there is no direct contact with live parts during operation. Attention: Incorrect polarity can lead to damage to the power supply and the connected devices. Note: Be sure to connect the polarity correctly. Attention: Continuous operation under full load can shorten the lifespan of the power supply. Note: Do not leave the power supply running at full load for long periods and allow regular breaks. Caution: Overheating can cause damage to the power supply and connected devices. Note:

Make sure the power supply is adequately ventilated during operation and monitor the temperature regularly. Caution: High temperatures can significantly shorten the lifespan of the power supply and connected devices. Note: Only operate the power supply within the recommended ambient temperatures (0°C to 40°C). Caution: Overheated components can cause fires. Note: Avoid use in direct sunlight or near heat sources. Caution: Overloading can lead to overheating and failure of the power supply. Note: Use the power supply only within the specified load limits. All power supplies have been developed in accordance with valid EMC regulations. These switching power supplies are rated as components and developed for installation in a terminal device. After integration into an end device, the EMC properties of the end system must be checked again.

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Introduction

The AZ-Delivery XL4016E1 Step-Down Module is a so-called buck converter or step-down converter. This DC/DC converter makes it possible to regulate the input voltage down to a desired value by means of a trimpot. The output voltage is always lower than the input voltage.

In normal operation, the converter can deliver an output current of 8A, with improved cooling even up to 12A.

Specifications

Input Voltage	5 to 40V
Output Voltage	1.2 to 36V (adjusted via trimmer)
Output Current	max 12A(rated 8A, peak 12A)
Output Power	100W (can reach 200W with enhanced cooling)
Switching frequency	300kHz
Efficiency	up to 95%
Dimensions	60x52x20mm
Weight	78g

Additional protective devices:

- overheat protection
- short circuit protection
- Fuse

Application example

In this example we use the XL-4016E1 Step Down Converter to drive two U 64 RGB LEDs from AZ-Delivery. The matrix is connected to the D1 Mini ESP32 running the WLED firmware. The current consumption of the matrix can be calculated as follows:

One RGB LED of the matrix requires 60mA at maximum brightness, there are 64 LEDs on the matrix and since we use two of them we have 128 LEDs in total.

$$0,06\text{A} \times 128 = 7,68\text{A}$$

The XL-4016E1 is connected to a suitable power supply. With the help of a voltmeter we can measure the output and set the correct output voltage with the trimmer, here 5V. After the correct voltage has been set, the connection with the matrix and the D1 Mini ESP32 can now be made, for this you should disconnect the converter from the power supply.

The output of the converter is connected to a mini breadboard, here we connect the D1 Mini ESP32.

XL-4016E1	D1 Mini ESP32
+	VCC
-	GND

After connecting the WiFi module, we can now connect the first matrix. Between the matrix and the converter, we connect an ElKo with 1000mF(ATTENTION: BE CAREFUL OF THE CORRECT POLARITY!).

XL-4016E1	Matrix 1
+	+5V(with ElKo)
-	GND(with ElKo)

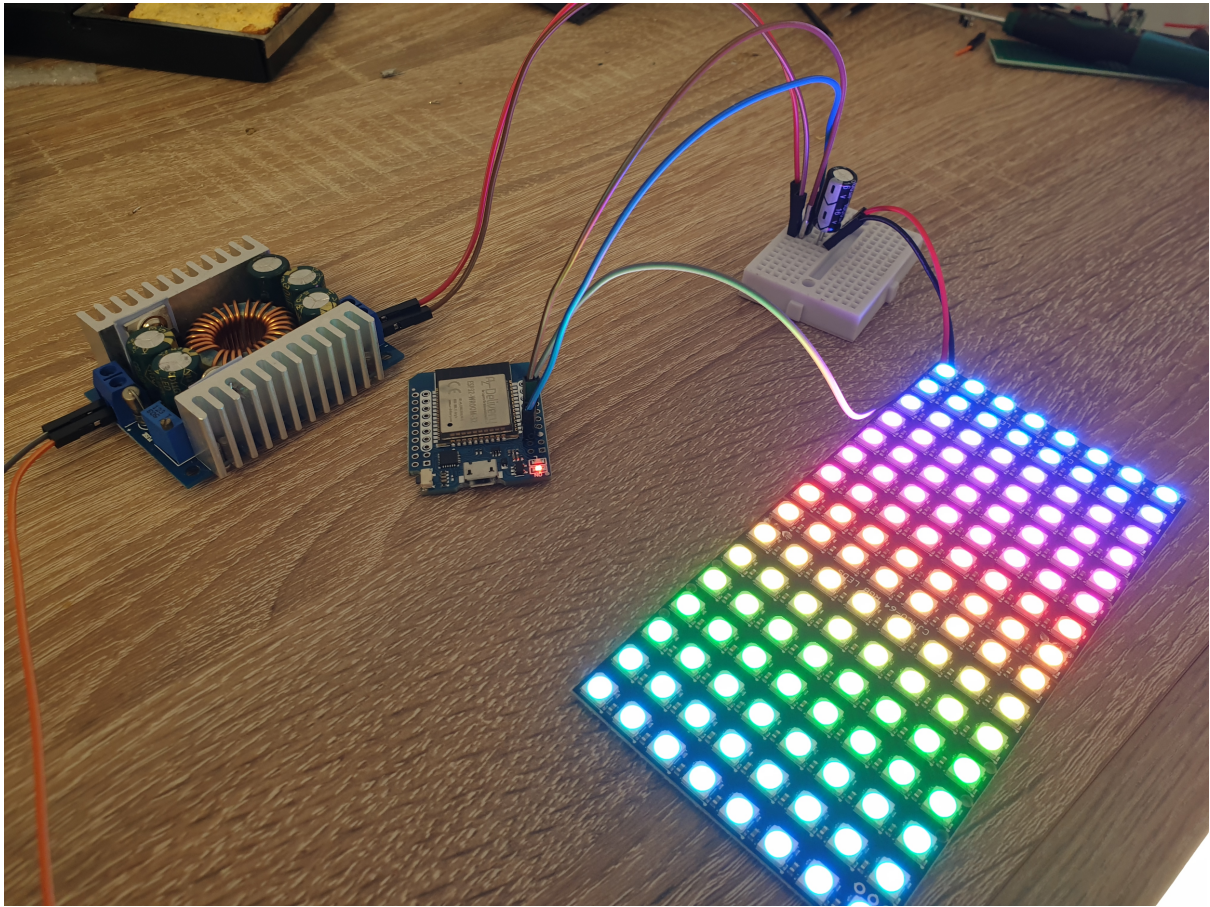
The D1 Mini ESP32 is connected to the first matrix, a resistor between 300 and 500 Ohm should be connected in between. We use a 470 Ohm resistor here.

D1 Mini ESP32	Matrix 1
GPIO16 (470 Ohm)	Din

Finally, we only have to connect matrix 1 with matrix 2.

Matrix 1	Matrix 2
Dout	Din
+5V	+5V
GND	GND

After everything has been connected, we can now switch on the power.



Now it's time to learn and create your own projects. You can do that with the help of many sample scripts and other tutorials that you can find on the Internet.

If you are looking for the high-quality products for Arduino and Raspberry Pi, AZ-Delivery Vertriebs GmbH is the right place for you. You will receive numerous application examples, complete installation instructions, eBooks, libraries and support from our technical experts.

<https://az-delivery.de>

Have Fun!

Impressum

<https://az-delivery.de/pages/about-us>